Probability: Formula Sheet

1. Definition and bounds

Classical:

$$P(A) = \frac{n(A)}{n(S)}$$

Complement: $P(A') = 1 - P(A)$
Range: $0 \le P(A) \le 1$
Exclusive and exhaustive: $\sum P(A_i) = 1$
Bounds:

$$P(A) + P(B) - 1 \le P(A \cap B) \le \min\{P(A), P(B)\}$$

2. Odds

In favour: $\dfrac{P}{1-P}$; against:

$$\frac{1-P}{P}$$

Odds in favour $p : q$:

$$P = \frac{p}{p+q}$$

Odds against $p : q$:

$$P = \frac{q}{p+q}$$

3. Dice and coins

n dice / coins: 6^n / 2^n outcomes
Two dice, sum s:

$$\frac{6 - |s-7|}{36}$$

r heads in n tosses:

$$\frac{{}^nC_r}{2^n}$$

4. Union and Venn diagram

Two: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Three:

$$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$$

n events: $S_1 - S_2 + S_3 - \cdots + (-1)^{n-1} S_n$
Only A: $P(A \cap B') = P(A) - P(A \cap B)$
Exactly one of two: $P(A) + P(B) - 2P(A \cap B)$
Neither: $P(A' \cap B') = 1 - P(A \cup B)$
Exactly two of three:

$$\sum P(A \cap B) - 3P(A \cap B \cap C)$$

At least two of three:

$$\sum P(A \cap B) - 2P(A \cap B \cap C)$$

Exactly one of three:

$$\sum P(A) - 2\sum P(A \cap B) + 3P(A \cap B \cap C)$$

5. Conditional and multiplication

Conditional:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Complement: $P(A'|B) = 1 - P(A|B)$
Subset $A \subseteq B$: $P(B|A) = 1$
Disjoint: $P(A|B) = 0$
Multiplication: $P(A \cap B) = P(A)P(B|A)$
Three:

$$P(A_1A_2A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1A_2)$$

6. Independent events

Test: $P(A \cap B) = P(A)P(B)$
Three: pairwise and $P(ABC) = P(A)P(B)P(C)$
At least one: $1 - q_1q_2 \cdots q_n$
Exactly one of two: $p_1q_2 + q_1p_2$
First success at trial k: $q^{k-1}p$
More than k trials: q^k
First of two in turn wins:

$$\frac{p}{1-q^2}$$

7. Total probability and Bayes

Total: $P(A) = \sum P(B_i)P(A|B_i)$
Bayes:

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum P(B_j)P(A|B_j)}$$

Extended: $P(C|A) = \sum P(B_i|A)P(C|B_iA)$

8. Binomial distribution

$B(n,p)$: $P(X = r) = {}^nC_r p^r q^{n-r}$
Ratio:

$$\frac{P(X=r)}{P(X=r-1)} = \frac{n-r+1}{r} \cdot \frac{p}{q}$$

Mean np, variance npq, SD $\sqrt{npq}$
Most probable: $(n+1)p - 1 \le r \le (n+1)p$
Least trials: $1 - q^n \ge k \iff q^n \le 1 - k$

9. Random variable

Distribution: $p_i \ge 0$, $\sum p_i = 1$
Mean: $\mu = \sum p_i x_i$
Variance: $\sigma^2 = \sum p_i x_i^2 - \mu^2$
Linear: $E(aX + b) = a\mu + b$, $\mathrm{Var}(aX + b) = a^2\sigma^2$
Sample without replacement:

$$P(X = r) = \frac{{}^aC_r \, {}^bC_{m-r}}{{}^{a+b}C_m}$$

10. Geometry and units digit

Geometrical:

$$P = \frac{\text{favourable measure}}{\text{total measure}}$$

Units digit $1, 3, 7, 9$:

$$\frac{4^n}{10^n}$$

Units digit $2, 4, 6, 8$:

$$\frac{8^n - 4^n}{10^n}$$

Units digit 5:

$$\frac{5^n - 4^n}{10^n}$$

Units digit 0:

$$\frac{10^n - 8^n - 5^n + 4^n}{10^n}$$